

Industrial Project Planning Report

Project 29:

Large Language Model Building for Bankwise Processes

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1. Objective

The goal of this project is to build a fine-tuned Large Language Models (LLMs) that focus on finance domains to perform retrieval-augmented generation (RAG) to assist bankwise processes. A prototype of will be demonstrated to illustrate the project outcomes. The comparison of various open-source LLMs on Retrieval Augmented Generation (RAG) for multilingual: Traditional Chinese, Simplified Chinese and English would also include in this project.

2. Motivation

Technologies such as LLM and generative AI have developed rapidly and are suitable for use in chat robots. The industry research team from bank of communication plans to first add the application of generative AI to its self-developed internal chatbot, Bell. RAG is the state-of-the-art technology that can improve the reliability and accuracy of the large language models. The financial and banking industry requires a strong domain specific knowledge and language comprehension. Fine-tuned LLMs combine the strengths of both information retrieval and generative models, allowing them to provide answers based on their existing and open-source knowledge, as well as up-to-date and private information from the banks. It supports bankers to get the significant information from large amounts of documents, improving operational efficiency and the quality of banking services.

3. Problem

However, most LLMs are built with a focus on the English language, resulting in limited proficiency in Chinese. This makes the fine-tuning process of LLM becomes more improvement, as it helps bridge the gap and empowers the Chinese finance industry to apply advanced AI technologies, as well as improve innovation and efficiency in an increasingly competitive landscape.

4. Proposed Solution

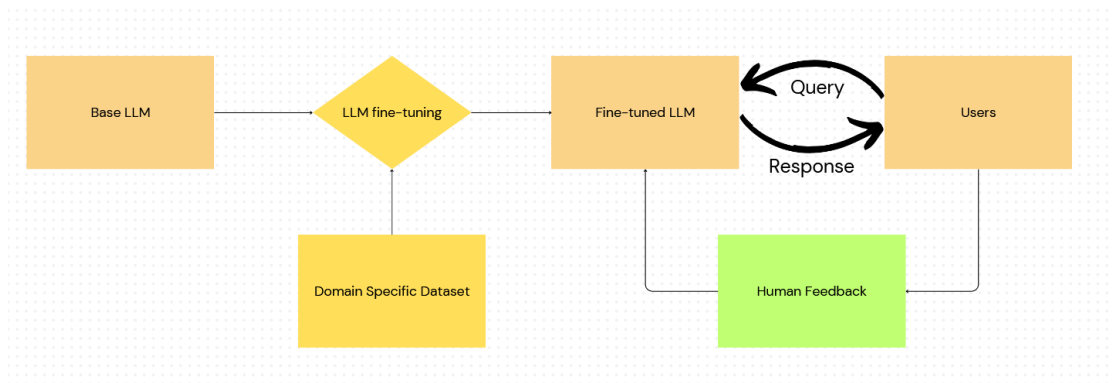


Figure 1: Architecture of LLM Building

4.1 Evaluation of open source LLMs

This project will begin with comparing the performance of various open-source LLMs, such as Llama and Qwen, on RAG tasks based on Chinese proficiency, accuracy, processing time, robustness. HKMA Supervisory Policy Manual would be used to generate the question-and-answer pairs to test their performance.

4.2 LLM Fine Tuning⁹

LLM with the best performance in RAG task in the evaluation would be chose to perform the fine-tune approach. More than 200 question-and-answer pairs will be created from datasets provided by the HKMA Supervisory Policy Manual to serve as the training set.

4.3 Feedback loop

The system in this project also contains a feedback loop to collect user feedback on the response's relevance and accuracy. The feedback is analysed to identify the gap of response and correct answer that helps to adjust the model. This approach ensures the system to meet user needs effectively and maintain high performance and quality.

5. Proposed timelines

Tasks	Due date
Evaluation on different LLMs	Mid-November
Fine-tune LLMs	End of 1 st semester
Other improvement: feedback loop / corrective RAG / self-reflective RAG	End of February
Prototype Deployment	End of March
Final Report	Mid-April